

NETWORKS AND PROTOCOLS

Exercises Sheet 2

Exercise 1

Assuming a classful addressing, determine the type of the following addresses (Network address, Host address, Broadcast address) and the class they belong to (A/B/C/D/E). Furthermore, for each class, determine the maximum number of hosts (in a given network), the mask and the broadcast address.

Address	Type	Class	Number of hosts	mask	broadcast address
130.192.0.0					
192.168.0.5					
80.255.255.255					
128.1.256.256					
134.188.255.255					
241.0.3.1					
224.0.0.3					

Exercise 2 (Subnetting using a classfull address (fixed mask))

Consider the address 129.15.0.0. We want to form 4 subnetworks using this address.

- How many bits do we need to meet our needs regarding the number of subnetworks?
- What is the ID of each subnetwork?
- What are the address and the mask of each subnetwork?
- What do you notice regarding the mask of the subnetworks?
- What is the broadcast address, starting address and ending address of each subnetwork?
- What is the maximum number of host addresses in each subnetwork?
- If the number of hosts in each subnetwork is as follows: subnet 1: 200 hosts; subnet 2: 5 hosts; subnet 3: 10000 hosts; subnet 4: 12000 hosts. What do you think about this subnetting method, is it an efficient one? What do you suggest as an alternative solution?

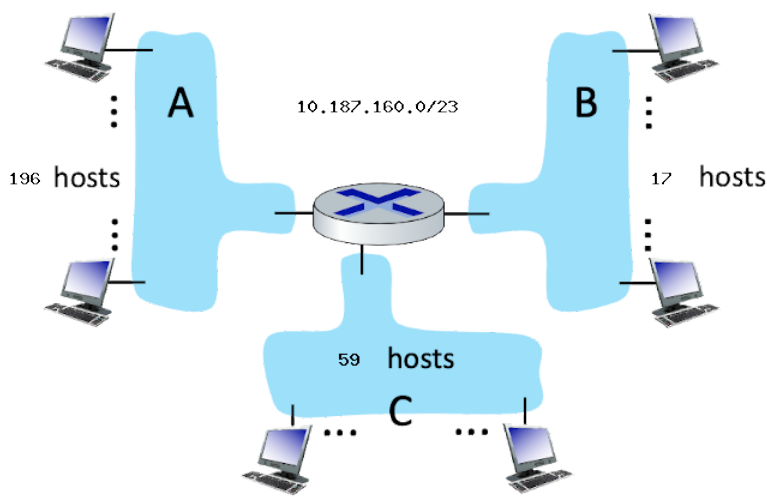


Figure 1: Exercise 3

Exercise 3 (Subnetting using a classless address (VLSM: Variable Length Subnet Masking))

Consider the router and the three attached subnets below (A, B, and C). The number of hosts is also shown below. The subnets share the 23 high-order bits of the address space: 10.187.160.0/23

Assign subnet addresses to each of the subnets (A, B, and C) so that the amount of address space assigned is minimal, and at the same time leaving the largest possible contiguous address space available for assignment if a new subnet were to be added. Then answer the questions below.

- Is the address space public or private?
- How many hosts can there be in this address space?
- What is the subnet address of subnet A? (CIDR notation)
- What is the broadcast address of subnet A?
- What is the starting address of subnet A?
- What is the ending address of subnet A?
- What is the subnet address of subnet B? (CIDR notation)
- What is the broadcast address of subnet B?
- What is the starting address of subnet B?
- What is the ending address of subnet B?
- What is the subnet address of subnet C? (CIDR notation)
- What is the broadcast address of subnet C?
- What is the starting address of subnet C?
- What is the ending address of subnet C?

Exercise 4 (Subnetting using a classless address (VLSM: Variable Length Subnet Masking))

Assuming a classless addressing plan, define the address ranges (in the form “network address/prefix length”) that can be used to handle a set of IP networks that include the number of hosts shown in the table below. The address spaces assigned to the networks should be assigned in order, one immediately following the other, within the address range 192.168.0.0/16. Determine also the broadcast address for each network.

Number of hosts	Network address / prefix length	Broadcast address
2		
27		
5		
100		
10		
300		
1010		
55		
167		
1540		

Exercise 5

Consider the network configuration described in the figure 2.

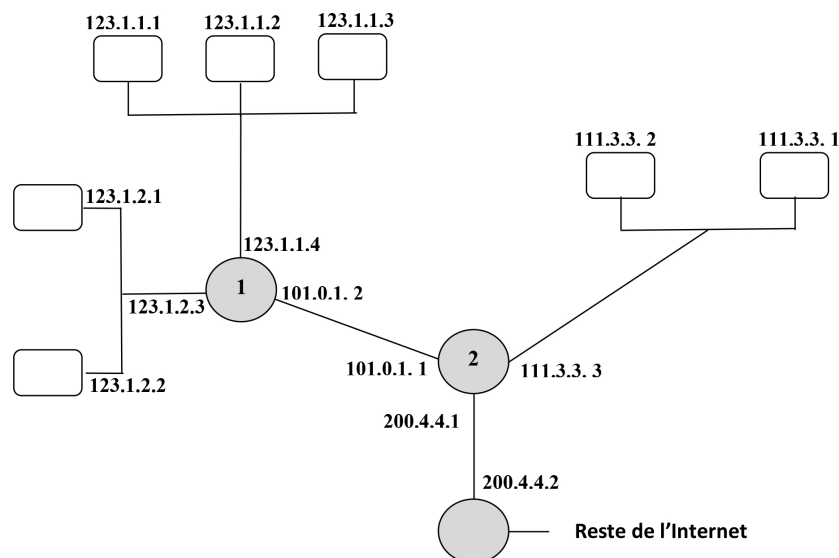


Figure 2: Network configuration.

What is the content of the routing table in each router (Router 1 and Router 2) by assuming that each subnet includes a block of 256 addresses?

Exercise 6

Given the network in figure 3, determine the routing table of R1 by aggregating the routes in a way such as:

- the addressing spaces are exactly equivalent to the original ones
- (or) there are the fewest possible entries in the routing table

Numbers in *italic* represent the cost of the link; assume unitary costs when not explicitly indicated in the figure.

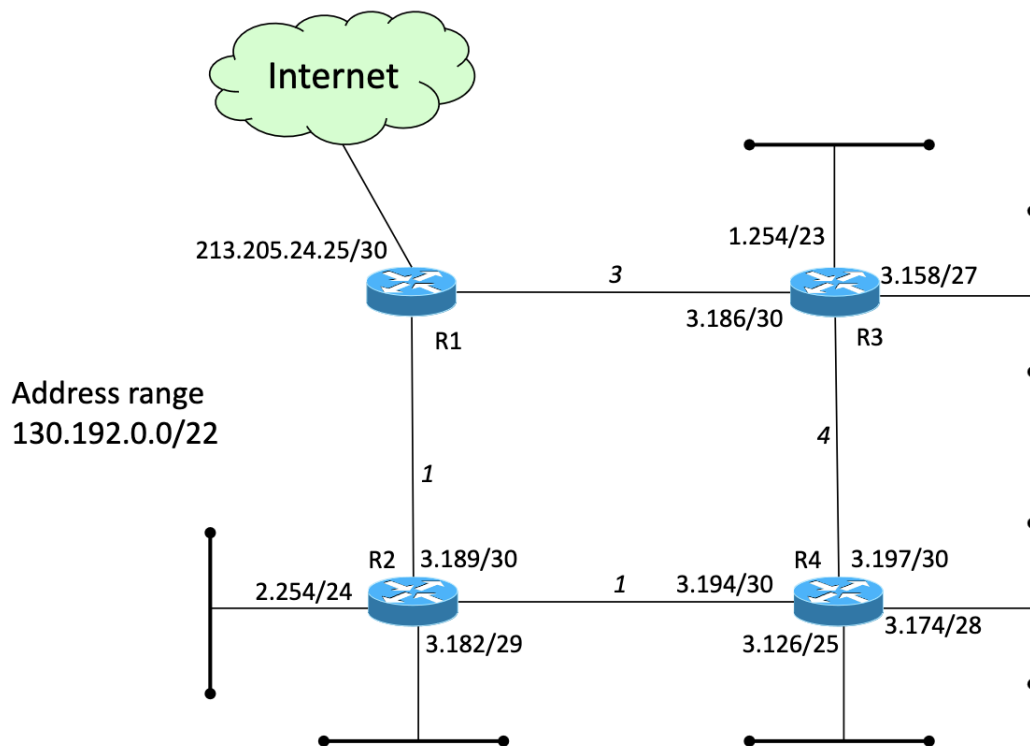


Figure 3: Network configuration.